Candidate Number ____________________ Fraction ____________________

2022 Republic of China Biology Olympiad National Team Selection Camp Practical Test Questions

Second **Practice** Session

Bioinformatics Practice—A

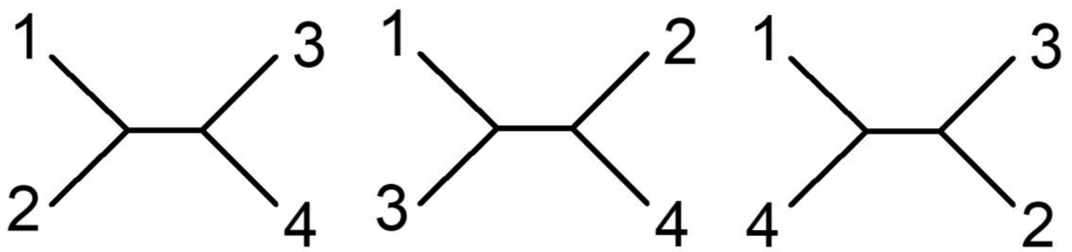
| project | quantity |
|---|---|
| Laptop (MEGA-11 program installed) | 1 unit |
| Laptop desktop - Folder containing Bioinformatics Practice A (inside) (Contains two files: heat shock protein 20.fas and heat shock protein 20.meg) | 1 |

Please note:

1. Please check if the exam number on the exam paper matches your exam number.
2. This exam paper (including the cover and the question paper) consists of 5 pages and must be returned in its entirety when submitting the exam.
3. This exam includes Practice A and Practice B, with a total time limit of 90 minutes.
4. Please answer the questions on this paper. You may write your answers on the back of the paper, but please indicate the question number.

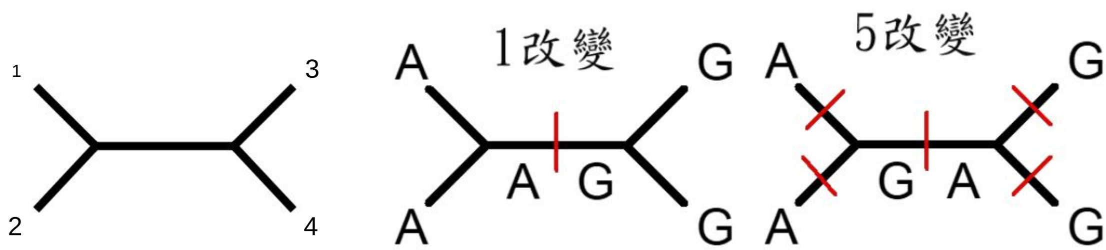
Question 1: A

kinship tree is a graphical representation used to show the evolutionary relationships of biological species. The evolutionary relationships constructed by a kinship tree are hypotheses; the branching patterns in the kinship tree show how species or other groups evolved from a series of common ancestors. Currently, most kinship trees are constructed based on molecular data, such as DNA or protein sequences. Generally, constructing a kinship tree requires four different steps: Step 1, identifying and obtaining a set of homologous DNA or protein sequences; Step 2, aligning these sequences; Step 3, estimating the tree structure from the aligned sequences; Step 4, presenting the kinship tree in a way that clearly expresses the relevant information. In terms of methodology for constructing kinship trees, one approach uses the maximum parsimony principle. This method is based on characteristic characters and infers the kinship tree by minimizing the total number of evolutionary steps required to interpret a given dataset distributed across the branches. When using DNA sequence data, the construction of a maximum parsimony tree is based on minimizing the number of changes in the genetic code. If we construct a maximum parsimony tree using the DNA sequences of four species, based on the possible branching relationships between species, we can infer three possible unrooted trees as follows:




Then, the bases that differ are labeled on the tree diagram, for example, a specific base in species 1 and 2 is...

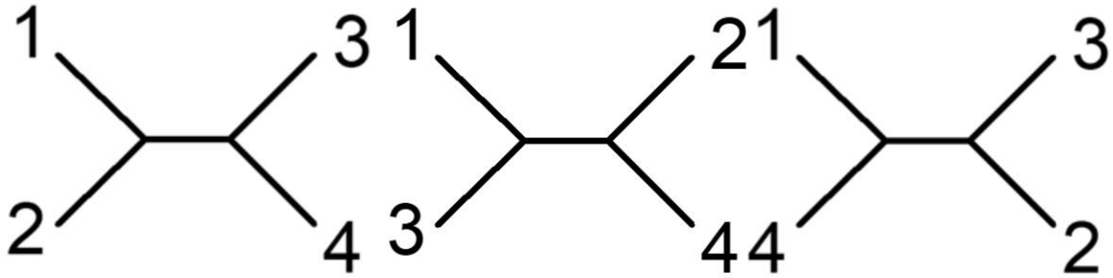
A. The base of species 3 and 4 is G. The rootless tree below may be changed by 1 or 5. According to the principle of maximum parsimony, the change by 1 is the most consistent with the principle of maximum parsimony.



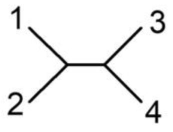
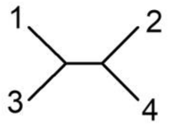
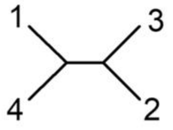

If we have a set of sequence data for 4 species, with a DNA sequence length of 10 bases, as follows:

| | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 |
|---|---|---|---|---|---|---|---|---|---|---|
| Species 1 | A | G | G | G | T | A | A | C | T | G |
| Species 2 | A | C | G | A | T | T | A | T | T | A |
| Species 3 | A | T | A | A | T | T | G | T | C | T |
| Species 4 | A | A | T | G | T | T | G | T | C | G |

Based on the above information, answer the following questions: 1. Using the principles and methods described above, construct the largest possible parsimony tree for each of the three possible kinship trees below, one base at a time. (3 points for each tree, 9 points in total)



2. Based on the principles and methods described above, please fill in the table below with the number of changes required for each base in the three possible phylogenetic trees. (0.5 points per cell, 15 points in total) Required Steps:

| Base 1, Base 2, Base 3, Base 4, Base 5, Base 6, Base 7, Base 8, Base 9, Base 10 | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|
|  | | | | | | | | | | |
|  | | | | | | | | | | |
|  | | | | | | | | | | |

3. Based on the above information, deduce which rootless tree constitutes the largest parsimonious relational tree. Please draw the tree structure below (2 points) and write down the number of base changes required (1 point).

4. Based on the maximum parsimony tree obtained above, deduce which bases have discriminative information. Please write down the base numbers? (2 points)

Question 2:

When organisms face heat stress, they activate heat shock protein genes to produce large quantities of heat shock proteins, which help maintain their balance.

Proteins fold normally; these proteins are found in many biological cells, and scientists have studied them in humans.

*(Homo sapiens), house mouse (Rattus), little house mouse (Mus* musculus), and ox (Bos *taurus) found a*

A heat shock protein with a molecular weight of 20 kDa was identified using molecular biology methods, and its gene fragments were found.

The segment, approximately 254 bases in length, is known to be a gene fragment containing an amino acid sequence, but the full sequence has not yet been obtained.

The file name for this sequence is heat shock protein 20.fas or heat shock protein 20.meg. Please refer to the file name provided.

Based on the given problem description, use the MEGA-11 program to perform the following analysis.

1. In the four 20kDa heat shock protein DNA sequences, the conserved nucleotide positions of this fragment are...

What percentage of the sites are nucleotides? What percentage of the sites are variable nucleotides?

Parsimony information: What percentage of nucleotide sites are represented? Singleton nucleotide sites.

What percentage does it represent? (1.5 points each, 6 points total)

| project | Answer |
|---|---|
| Conserved(C): | |
| Mutation (Variable)(V): | |
| Parsimony information(P): | |
| Singleton(S): | |

2. Among the four 20kDa heat shock protein DNA sequences, because they are known to contain amino acid sequences...

Due to the fragment, please use the relevant functions of the MEGA-11 program to translate the DNA sequence into possible amino groups.

Please fill in the acid sequence and the first 10 amino acid sequences in the table below (0.375 points per cell, 15 points in total).

| Species 1 | | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 |
|---|---|---|---|---|---|---|---|---|---|---|
| people | | | | | | | | | | |
| House mice | | | | | | | | | | |
| House mouse | | | | | | | | | | |
| cattle | | | | | | | | | | |

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Second **Practice** Session

Bioinformatics Practice—B

| | Quantity: |
|---|---|
| Project | 1 unit |
| Laptop Desktop - Folder for Bioinformatics Practice **(B )**<br>Q1 Folder: NCBI<br>Vectorbuilder<br>Q1_sequence.txt<br>Q2 Folder: BioEdit<br>Expasy<br>Reverse Complement<br>Q2-1_sequence.txt<br>Q2-2_sequence.txt<br>Q2-3_sequence.txt | 1 |

ÿ Please note: 1.

Please verify that the exam number on the exam paper matches your exam number. 2. This exam paper (including cover and question paper) consists of 5 pages and must be returned in its entirety upon submission. 3. This exam includes both Practical Exercise A and Practical Exercise B, with a total time limit of 90 minutes. 4. Please answer the questions on this exam paper. Answers may be written on the back of the exam paper, but please indicate the question number. 5. Please note that only the websites and bioinformatics software provided in the folder are permitted during the exam. All other software or websites are strictly prohibited and will result in point deductions if discovered.

Question 1: NCBI-BLAST (Basic Local Alignment Search Tool) is a gene sequence alignment tool that compares DNA or protein sequences with sequences in the NCBI database to obtain similar sequences and identify the species to which the sequence belongs. The ratio of guanine (G) to cytosine (C) in the whole sequence is called GC content. GC content affects DNA stability and mRNA secondary structure, etc. Vectorbuilder can be used to calculate the GC content and length of a sequence. Recently, researchers obtained an unknown DNA sequence, Q1_sequence.txt, and want to know the species from which this sequence comes, its GC content, and its sequence length. Please use NCBI and Vectorbuilder in the Q1 folder to analyze Q1_sequence.txt and answer the following questions.

Question 1: Using NCBI-BLAST sequence alignment to identify the species, what species does the sequence in Q1_sequence.txt belong to (please write the scientific name)? (5 points)

Question 2: Following on from the previous question, to which kingdom in biological taxonomy does this species belong? (5 points)

Question 3: Using Vectorbuilder, analyze the GC content and base pairs (bp) of the Q1_sequence.txt sequence (Window size 50). (4 points)

Question 4. Following on from the previous question, (1) what is the range of bases with the highest GC content? (2) What is the range of bases with the lowest GC content? (In units of 1000 bases) (4 points)

Question 2: Researchers amplified a DNA fragment of a certain species using PCR technology and then sequenced it to obtain Q2-1_sequence.txt, Q2-2_sequence.txt, and Q2-3_sequence.txt. Q2-1_sequence.txt is the coding sequence (CDS) of the wild type; Q2-2_sequence.txt and Q2-3_sequence.txt are the coding sequences of the mutant type. The sequence orientation of Q2-1_sequence.txt and Q2-3_sequence.txt is from the 5' end to the 3' end, while the sequence orientation of Q2-2_sequence.txt is from the 3' end to the 5' end. Researchers first used Reverse Complement to change the sequence orientation, then used BioEdit to align the nucleic acid sequences and identify the mutation sites. Next, they used ExPASy to translate the DNA sequence into amino acids. Finally, they used BioEdit to determine whether the amino acid sequence mutation site was a nonsense mutation, a missense mutation, or a silent mutation. Students are asked to analyze the three sequences Q2-1_sequence.txt, Q2-2_sequence.txt, and Q2-3_sequence.txt in the Q2 folder using Reverse Complement, ExPASy, and BioEdit, and answer the following questions.

Question 5: Perform nucleic acid sequence alignment on three sequences and list their variant sites (answer presented as: site 10, AÿT). (16 points)

Question 6: Following on from the previous question, perform amino acid sequence alignment on the three sequences. Describe the type of mutation at a specific amino acid mutation site. (Answer presented as: MÿL at the 10th site; type of mutation) Method? (16 points)